Sean Colsen *Senior Full Stack Software Engineer*

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Technical Skills

Proficient TypeScript, JavaScript, Svelte, React, Tailwind, Python, Django, PostgreSQL, Redis, MySQL, SQLite, PL/pgSQL, SQL, Rust, DuckDB, Node.js, vite, Docker, Playwright, Jest, pytest, HTML, CSS, SVG, PWA, MkDocs, GitHub

Exposure Axum, Vue, Nuxt, PHP, Linux, Bash, Storybook, a11y, pgTAP, mypy, Selenium, AngularJS, Flask, Symfony, AWS, GCP, Ansible, Kubernetes, Nginx, Ruby, C++, Java, Mathematica

Experience

Senior Software Engineer for Mathesar Foundation 2021 - 2026

- Led front end development of open source spreadsheet and database hybrid app in TypeScript and Svelte, giving non-technical users an interactive UI to view, edit, and query PostgreSQL data, **achieving 5K GitHub stars**
- Owned and architected key feature which **drove $2 million in funding** by creating templating system enabling users to define custom record summaries that appear in foreign key cells, using TypeScript, Svelte, Python, and PL/pgSQL
- Engineered cell selection and clipboard system for spreadsheet UI in TypeScript and Svelte, letting users **move data 40X faster**
- **Led team of 5 engineers and 1 designer** through two full releases during founder's 3-month sabbatical
- Designed and implemented file attachment UI in TypeScript and Svelte, **used by 300+ developers and organizations**
- Managed and triaged all public activity on Mathesar's GitHub repository using semi-automated system built with GitHub's API, SQL, and DuckDB, enabling **over 30X faster responses** to users

Software Engineer for Insurance Toolkits 2021

- Developed custom web-scraping system using Python and Selenium to **collect 500,000+ rows of insurance rate data from 10+ carriers across 40 states**, enabling launch of new B2B SaaS insurance sales agent product
- Re-architected front end of life insurance sales tool from JavaScript and Vue.js to TypeScript and Nuxt, **enabling expansion from final expense into term and other product lines**
- Designed and built back end features and REST APIs in Django, PostgreSQL, and Redis, type-checked with mypy
- Found and fixed critical security vulnerabilities through proactive code review, including broken access control that **let any user take over an admin account** and XSS via user-uploaded files

Software Engineer for Left Join Labs 2016 - 2020

- Founded and ran consultancy developing and supporting CiviCRM, an open source CRM, for nonprofits
- Contributed **1,200+ commits and 100+ merged PRs** to CiviCRM's code and docs in PHP, JavaScript, AngularJS, and MySQL
- Reviewed **250+ pull requests** and created a PR review checklist **adopted into CiviCRM's official review standards**
- Served on CiviCRM's 6-person Security Team, fixing **20+ vulnerabilities**, triaging private reports, and planning security releases
- Led CiviCRM Documentation Working Group, building custom publishing system for docs.civicrm.org with PHP, Symfony, and MkDocs, and **migrating 10 years of documentation** into it from Confluence

Web Developer & IT Manager for Bikes Not Bombs 2008 - 2016

- Ran all IT as a one-person department for this nonprofit of 20+ staff, managing systems, data, and software
- Analyzed decades of donor history with SQL to prioritize fundraising appeals, **increasing appeal revenue 1.6X**
- Migrated **40K contacts** into CiviCRM and built custom PHP and MySQL extensions, supporting growth to **80K+ contacts**

Math Content Developer for Pearson 2006 - 2008

- Developed **1,000+ interactive math tutorials** for college-level calculus/trigonometry students in Pearson's MathXL platform

Project Work

RadioCrate 2026 - current React, TypeScript, Tailwind, Rust, Axum, DuckDB, PWA
High-performance music manager and mobile player with user-defined data fields and querying via Querydown

Querydown 2023 - current Rust, SQL
Expressive **query language** for relational databases that compiles to SQL

Octave Compass 2020 Svelte, TypeScript, SVG, PWA
Interactive tool for musicians to explore **1,000+ named scales** and their chords, with 70+ Svelte components

Education

Bachelor of Science in **Electrical Engineering**, Northeastern University, Boston